Module 9 Lab 1: Textbook Problem 6.1, in the Cloud

Course: CS 82A Introduction to Data Science
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Colab URL: https://colab.research.google.com/drive/1RLH1vDkW6uex-LsARU-e6fPPJmvtK4ys?usp=sharing
Repo URL: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module9/module9_lab1.ipynb

AI disclosure: Used Claude (Anthropic) to verify the code was correct and explain concepts.

Cloud check

This cell shows where the notebook is running. In Colab it prints `True`.

```
import sys
import platform

print("Running in Google Colab:", "google.colab" in sys.modules)
print("Python version:", platform.python_version())
print("Operating system:", platform.platform())
```

```
Running in Google Colab: True
Python version: 3.13.16
Operating system: Linux-6.6.122+-x86_64-with-glibc2.39
```

Problem 6.1: Pounds to kilograms

The conversion factor is 1 lb = 0.453592 kg.

```
# Function that converts pounds (lbs) to kilograms (kgs)
def lbs_to_kgs(lbs):
    """Return the weight in kilograms for a weight given in pounds."""
    return lbs * 0.453592
```

Calling the function

```
# Call 1: one pound
print("1 lb =", round(lbs_to_kgs(1), 4), "kg")

# Call 2: one hundred pounds
print("100 lbs =", round(lbs_to_kgs(100), 4), "kg")

# Call 3: my own weight in pounds
my_weight_lbs = 118
print(my_weight_lbs, "lbs =", round(lbs_to_kgs(my_weight_lbs), 4), "kg")
```

```
1 lb = 0.4536 kg
100 lbs = 45.3592 kg
118 lbs = 53.5239 kg
```

```
# Summary table of all three calls
import pandas as pd

weights_lbs = [1, 100, my_weight_lbs]
table = pd.DataFrame({
    "Pounds (lbs)": weights_lbs,
    "Kilograms (kgs)": [round(lbs_to_kgs(w), 4) for w in weights_lbs],
})
table
```

	Pounds (lbs)	Kilograms (kgs)
0	1	0.4536
1	100	45.3592
2	118	53.5239

Next steps: Generate code with `table` New interactive sheet